

Red eye cornea and uveitis

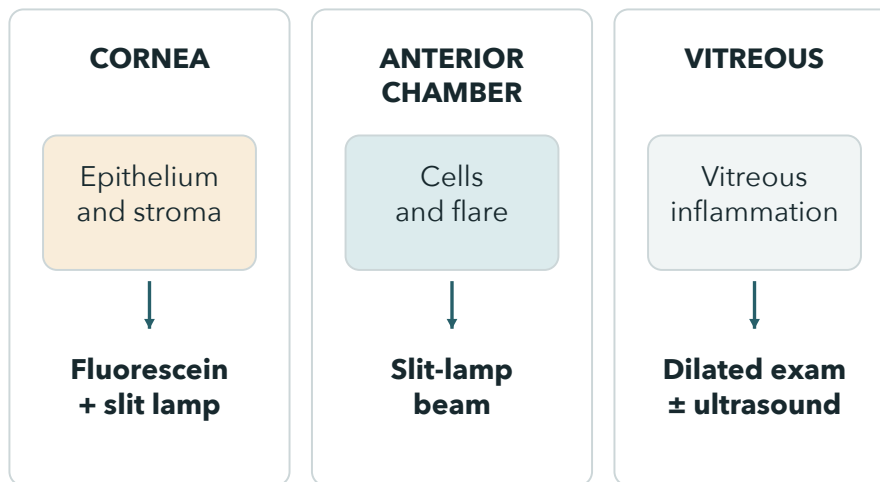
M3-M5 | Localise the problem, identify emergencies, choose the next step

Triage the red eye

Start with the risk

Reduced vision, significant pain or photophobia, contact lens wear, recent surgery, chemical exposure or trauma changes the urgency. Record visual acuity in each eye; a reassuring acuity alone does not exclude serious disease.

Localise the inflammation



Original schematic | Corneal defect, anterior chamber cells and vitritis identify different compartments; more than one may be involved.

Clues that change the differential

Pattern	Priority diagnosis / action
Infiltrate + epithelial defect	Microbial keratitis. Same-day corneal assessment.
Cells + ciliary flush + photophobia	Anterior uveitis. Slit-lamp assessment and infection screen.
Haze + shallow chamber + very high IOP	Acute angle closure. Immediate pressure-lowering pathway.
Postoperative deterioration + vitritis	Endophthalmitis. Emergency ophthalmic treatment.

Focused examination

Inspect lids, cornea and fluorescein staining; assess pupils and anterior chamber. Measure IOP only when safe. Chemical irrigation comes before a lengthy examination; suspected open globe needs a rigid shield, no pressure or tonometry, and emergency referral.

References College of Optometrists | Urgency of referral | College of Optometrists | Microbial keratitis, 2024 | College of Optometrists | Postoperative endophthalmitis

Microbial keratitis: sample and treat

Case pivot

A contact lens wearer has a central stromal infiltrate, an overlying epithelial defect and falling acuity. Treat this as a sight-threatening corneal infection, even if the discharge resembles conjunctivitis.

History directs the microbiology

Exposure / clue	What it raises
Overnight lens wear	Bacteria, including <i>Pseudomonas</i> . Stop lenses.
Lens use + water exposure	<i>Acanthamoeba</i> ; pain may be disproportionate.
Vegetative trauma; feathery edge / satellites	Fungal infection. Early specialist sampling matters.
Dendrite + reduced sensation	HSV epithelial disease; use the antiviral pathway.

Next steps in a vision-threatening ulcer

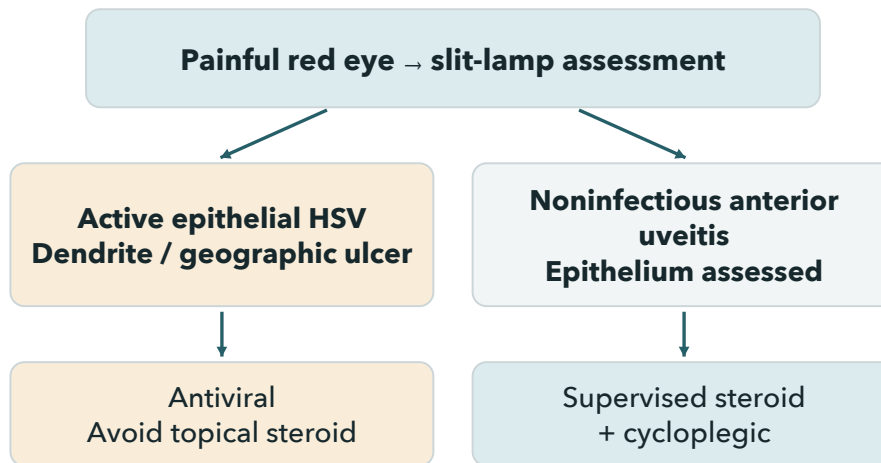
- 1 Same-day ophthalmology. Document size, position, depth, acuity, thinning and anterior chamber reaction.
- 2 Scrape for microscopy and culture when large, central, deep, atypical or failing treatment. Obtain samples promptly if feasible.
- 3 Start intensive empirical topical antimicrobials immediately after sampling. Do not wait for culture results; use local protocols.
- 4 Review closely and adapt to organism, susceptibility and response. Admit if severe or drops cannot be delivered reliably.

Steroids and safety

No unsupervised steroid or patch over a suspected infection. A corneal specialist may add steroid in selected responding bacterial ulcers; fungal and *Acanthamoeba* infection require different treatment. Suspected thinning or perforation requires urgent protection and surgical assessment.

Uveitis and the HSV steroid decision

Examine before choosing a steroid



Original decision map | Infection, corneal epithelium and IOP determine treatment, not redness alone.

The comparison that prevents harm

Finding	Clinical implication
Dendrite / geographic epithelial ulcer; reduced sensation	Active epithelial HSV. Antiviral treatment; avoid topical steroid on active epithelial disease.
Cells / flare, keratic precipitates, small or irregular pupil	Anterior uveitis. Ophthalmologist-led steroid and cycloplegic if noninfectious.
Unilateral uveitis + high IOP / sectoral iris atrophy	Consider viral uveitis. Antiviral plus inflammation control may be required.

Treat and monitor

For noninfectious anterior uveitis, topical steroid controls inflammation; a cycloplegic relieves ciliary spasm and limits posterior synechiae. Dose and taper follow examination findings. Review acuity, cells, IOP and complications, including cataract and cystoid macular oedema.

Ask beyond the eye

Recurrent disease, inflammatory back pain, psoriasis or bowel symptoms raise HLA-B27-associated disease. Granulomatous or bilateral inflammation raises sarcoidosis and infectious causes. Ophthalmology selects investigations, including syphilis testing and targeted HLA-B27 or chest imaging; avoid an indiscriminate laboratory panel.

The deteriorating postoperative eye

Case pivot

Vision improves after cataract surgery, then deteriorates with redness, pain, hypopyon and vitritis. Infection must be considered immediately; absence of pain does not provide reassurance.

Infection vs toxic anterior segment syndrome (TASS)

Feature	Endophthalmitis	TASS
Typical context	Often days after surgery; may occur earlier or later	Often within 12-48 hours
Posterior segment	Vitritis, poor red reflex / fundal view	Predominantly anterior inflammation
Symptoms	Marked visual decline; pain is variable	Blurred vision; often less pain
Action	Urgent sampling + intravitreal antimicrobials	Specialist anti-inflammatory treatment once infection excluded

Act on suspected endophthalmitis

- 1 Immediate ophthalmic / vitreoretinal assessment. Check acuity, wound, anterior chamber and vitreous.
- 2 If the fundus is obscured, ophthalmic ultrasound can assess the posterior segment while treatment is arranged.
- 3 Obtain intraocular samples and give empirical intravitreal antibiotics promptly, usually vancomycin plus ceftazidime under local protocols. Culture guides subsequent treatment.
- 4 Assess the need for vitrectomy and follow response closely. Topical or systemic antibiotics alone are inadequate for acute postoperative bacterial infection.

Avoid the timing trap

Timing overlaps. A severe unexplained postoperative inflammatory eye should be managed as possible infection until assessed. Steroid escalation alone can delay the treatment that preserves vision.

Chemical injury: stop continuing exposure

The examination follows decontamination

Start copious irrigation immediately with available clean water or saline; do not delay for a full history or specialist fluid. Remove contact lenses and avoid chemical neutralisation with an opposing acid or alkali.

The pH loop

- 1 Check surface pH in both eyes when feasible, without delaying irrigation.
- 2 Irrigate; evert the lids and inspect / clear particulate chemical from the fornices.
- 3 Pause and recheck pH after at least five minutes. A rebound suggests retained material or continuing chemical release.
- 4 Repeat irrigation and particle removal until pH remains physiological; arrange urgent ophthalmic assessment.

After pH stabilises

Assess	Why it matters
Acuity; epithelial defect	Baseline vision and surface damage
Corneal clarity; limbal ischaemia	Severity and prognosis; limbal injury threatens epithelial renewal
Conjunctiva / fornices	Residual particles and tissue injury
IOP; anterior chamber	Pressure complications and deeper injury

Clinical discussion

A normal pH immediately after washing is not the endpoint if it rebounds. Subsequent antibiotic, cycloplegic, lubrication and supervised anti-inflammatory treatment depend on injury severity; severe burns require specialist follow-up for melting, limbal failure and glaucoma.

Reference map

Assessment: referral guidance. Infection: microbial keratitis and bacterial keratitis guidance. Steroid decision: HSV and anterior uveitis. Postoperative pathway: endophthalmitis. This page: chemical injury guidance. All diagrams are original teaching schematics.